Improving the Basic Model

BZAN 554 Group Assignment 2 · Group 6

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2026-10-29

How to read the comparisons

- Keep earlier additions; report gains and regressions.
- SKU dropout stays at **50%** after introduction.
- All tests train for **up to 40 epochs**; early stopping may end runs sooner.
- Step tables: same split, seed 42, batch 1,024.
- Rows: **349,975 train / 74,736 validation / 75,289 test**.
- Fit preprocessing on training data; select by validation.
- Early steps use final weights; step 6 adds best-checkpoint selection.
- Final confirmation: **three seeds** with the same training budget.

0. Baseline

Three sigmoid hidden layers: **256 / 128 / 64** neurons. SKU/category embeddings, standardized raw inputs, linear output, and Adam at 0.001.

Model	Validation R^2	Validation RMSE
Basic model	0.3636	39.69

No added feature engineering, dropout, L2, clipping, rate reductions, early stopping, or best-epoch selection.

1. Add log features

Add log1p versions of price, order, and duration alongside the raw inputs. Fit scaling on training data only.

Model	Validation R^2	Validation RMSE
Before addition	0.3636	39.69
After addition	0.3336	40.62

Change: R^2 -0.0299, RMSE +0.92.

2. Add square-root features

Add square-root versions of price, order, and duration. The model now has nine numeric inputs.

Model	Validation R^2	Validation RMSE
Before addition	0.3336	40.62
After addition	0.3389	40.46

Change: R^2 +0.0053, RMSE -0.16.

3. Add hidden dropout

Drop **5%** of activations after each hidden layer during training. Keep this rate in every later model.

Model	Validation R^2	Validation RMSE
Before addition	0.3389	40.46
After addition	0.4046	38.39

Change: R^2 +0.0657, RMSE -2.06.

4. Add SKU dropout

Drop the whole SKU embedding for **50%** of training examples. Keep **50%** in every later model.

Model	Validation R^2	Validation RMSE
Before addition	0.4046	38.39
After addition	0.5837	32.10

Change: R^2 +0.1791, RMSE -6.29.

5. Add L2 penalties

Add a **0.0001 L2 penalty** to hidden-layer kernels. Evaluation RMSE excludes the penalty.

Model	Validation R^2	Validation RMSE
Before addition	0.5837	32.10
After addition	0.5996	31.49

Change: R^2 +0.0159, RMSE -0.62.

6. Add validation-based training controls

Keep the best validation checkpoint: **epoch 15**. This combined step also includes norm clipping at 10, learning-rate reductions, and early stopping with patience 12.

Model	Validation R^2	Validation RMSE
Before addition	0.5996	31.49
After addition	0.6177	30.77

Change: R^2 +0.0181, RMSE -0.72.

7. Switch Adam to RMSprop

Switch **Adam to RMSprop**, keeping the initial learning rate at **0.001** and all other settings fixed.

Model	Validation R^2	Validation RMSE
Before addition	0.6177	30.77
After addition	0.6210	30.63

Change: R^2 +0.0033, RMSE -0.13.

8. Reduce the L2 penalty

Lower the hidden-kernel L2 coefficient from **0.0001 to 0.00001**. Keep RMSprop and the previous settings.

Model	Validation R^2	Validation RMSE
Before addition	0.6210	30.63
After addition	0.6217	30.60

Change: R^2 +0.0007, RMSE -0.03.

Confirm the improvement across three seeds

Compare seed 42 with the three-seed average.

Configuration	Seed 42: R^2	3-seed mean: R^2
Adam + L2 0.0001 (step 6)	0.6177	0.6170
RMSprop + L2 0.00001 (step 8)	0.6217	0.6222

Matched confirmation gain: mean R^2 **+0.0053**, positive in **all three seeds**.

The seed-42 scores continue directly from steps 6 and 8. The right column averages seeds 42, 101, and 202.

Final model: distinguish the reported scores

Chosen configuration: RMSprop + L2 0.00001 (step 8).

Packaged checkpoint: seed 202, epoch 26; selected using validation.

Reported score	R^2	RMSE	MAE
Three-seed mean validation	0.6222	30.58	–
Selected checkpoint: validation	0.6235	30.53	12.51
Selected checkpoint: test	0.6245	31.40	12.74

The mean summarizes repeatability across seeds. The other rows evaluate the saved checkpoint on validation and test data.

Other experiments: changes that did not help

Each test starts from the same RMSprop model.

Tested change	Before R^2	After R^2	Change
Seven sigmoid layers	0.6210	≈ 0	-0.6210
L2 penalty 0.001	0.6210	0.5981	-0.0229
Batch size 4,096	0.6210	0.6194	-0.0016
Batch norm before activation	0.6210	0.6156	-0.0054

Matched parent: RMSprop with L2 0.0001, validation R^2 **0.6210**. Each row changes one setting from this model; the changes do not accumulate. Same split, seed 42, and training budget.

Evaluation

- RMSE uses supplied quantity units and excludes L2 penalties.
- Validation selects settings; test scoring follows frozen selection. The test set has historical exposure.
- Predictions describe completed selling episodes, not advance forecasts.
- Results apply to this setup and budget; all trials remain logged.

Exact tuning values and whole-SKU masking are project choices.

Team contributions

Team member	Contribution
Ian Allish	[To be filled in]
Joseph Falcone	[To be filled in]
Will Schneider	[To be filled in]
Noah Hoogerheide	[To be filled in]

AI assistance: Codex implemented the experiments and logged the results